[PDF Slides] [HTML Slides] [Colab Notebook]

Course Overview

Welcome to Introduction to Machine Learning! This course will provide you with a comprehensive understanding of machine learning concepts, algorithms, and practical applications.

Course Structure

- 10 sessions of 4 hours each
- Combination of theory and practice
- Hands-on exercises using Python
- Interactive elements (polls, cloud numbers)

Learning Objectives

By the end of this course, you will:

- Understand fundamental ML concepts
- Be able to implement basic ML algorithms
- Know how to evaluate ML models
- Have practical experience with real-world datasets

{% include _snippets/ml/what-is-ml.md %}

{% include _snippets/ml/history.md %}

{% include _snippets/ml/applications.md %}

Benefits and Risks

Benefits

- Automation of complex tasks
- Improved decision-making
- Personalization
- Efficiency gains

Risks and Challenges

- Data privacy concerns
- Algorithmic bias
- Job displacement
- Ethical considerations

Tools for ML Implementation

Python Ecosystem

1. Core Libraries
    - NumPy: Numerical computing
    - Pandas: Data manipulation
    - Scikit-learn: ML algorithms
    - TensorFlow/PyTorch: Deep learning

2. Development Tools
    - Jupyter Notebooks
    - Google Colab
    - VS Code with Python extensions

Practical Session

In this session's practical component, we will:

1. Set up our development environment
2. Explore basic Python libraries for ML
3. Create our first ML pipeline

Resources

Recommended Reading

- "Hands-On Machine Learning with Scikit-Learn and TensorFlow" by Aurélien Géron
- "Python for Data Analysis" by Wes McKinney
- Online courses and tutorials

Additional Materials

- Course GitHub repository
- Discussion forum
- Office hours schedule

Next Session Preview

In our next session, we will dive deeper into:

- Data preprocessing techniques
- Feature engineering
- Basic ML algorithms
- Model evaluation metrics